

Dexter Fernandes

SLAM Computer Vision Engineer

Fluent in English, Hindi, Konkani

Bristol, EN | <https://www.linkedin.com/in/dexterferns/> | +44 7436007369 | dexterfernandes11@gmail.com

PROFESSIONAL SUMMARY

Computer Vision Engineer with 3+ years of experience in low-latency vision solutions for Tier 1 companies, including a construction firm in Texas and a UK oil and gas company in Oman. Pursuing a Master's in Robotics, focusing on Visual SLAM and sensor fusion for an **Autonomous Radiological Survey Robot**. Skilled in collaborating with multinational clients, applying state-of-the-art research in fast-paced start-up environments, and deploying AI solutions to production including testing, QA, and monitoring.

SKILLS

Deep Learning | Computer Vision | Time-series Analysis | Machine Learning | Test Driven development(TDD) | Version Control | Containerisation | Leadership and Management | Client Interaction | MLOps | Distributed Event Data-Streaming | Project Management | Simultaneous Localisation and Mapping | Troubleshooting | Problem-solving

WORK EXPERIENCE

Esbaar

AI Engineer

Muscat, Oman

January 2023 – August 2023

Z-Alert

- Developed custom time-series algorithms in Keras and PyTorch to classify 15 oil rig activities, managing and analysing 3.6 million data points from numerous sensors.
- Streamlined hyperparameter tuning for time-series analysis models through MLOps integration, orchestrating the training of approximately 500 time-series models within a 48-hour period – improving F1-score by 39%.
- Led the team that adapted YOLOv5 to detect rig workers and heavy equipment, fostering a safer work environment, improving model F1-score by 17% through model maintenance, iteration, and drift management.
- Collaborated with multinational clients to address technical challenges, achieving key milestones and delivering solutions within 4 months, while gaining valuable insights into cross-cultural communication and project management.
- Leveraged Apache Kafka with Docker containers to re-factor a monolithic application into a microservices architecture, reducing software update intervals and improving customer satisfaction.
- Eliminated a persistent, year-old segmentation fault in a massive monolithic application in just one day, showcasing exceptional problem-solving skills and technical prowess.
- Engineered an accessible MLOps pipeline that streamlined and automated the training of over 300 models, dramatically boosting efficiency and scalability while ensuring full visibility.

LivNSense Technologies Pvt. Ltd.

Lead Software Engineer - AI

Bengaluru, India

May 2020 – December 2022

ViCAS – Safety-based Anti-Collision Early Detection System

- Managed a team of 2-4 persons that contributed to every part of the pipeline, which included curating the training and testing datasets, managing model life cycle in production, and tackling edge cases.
- Transformed vision model inference speed with a remarkable 3.6x increase on the same hardware, leveraging advanced optimization techniques and deep learning expertise to significantly enhance performance and efficiency.
- Accelerated model training by reducing time from 2.5 days to 4 hours on the same hardware, highlighting a deep understanding of deep learning frameworks and advanced optimization capabilities.
- Enhanced the production inference application by writing modern C++ code and facilitating its porting from Python to improve inference times and enhance application security.
- Overhauled the labelling platform and repurposed older production models for auto-labelling, increasing annotator output from 600 to 3000 images per day and maximising resource efficiency.
- Surpassed the existing vision solution by revamping the dataset, re-formulating the problem statement, and reducing inference time from 250 ms to 90 ms on the same edge hardware.
- Systematically resolved significant model drift in a single day by constructing and training on a synthetic dataset mimicking production changes, reducing false positives by 50% and reassuring clients until raw data was procured.
- Represented the company as a consultant abroad, understanding client requirements and delivering a pilot solution with real-time analytics within two weeks. – Workplace Hazard Detection System

EDUCATION

University of Bristol

MSc in Robotics (Ongoing)

Bristol, England

September 2023 – September 2024

Visvesvaraya National Institute of Technology

B. Tech in Electrical and Electronics Engineering

Nagpur, India

July 2016 – May 2020

SOFTWARE STACK

Python | C++ | OpenCV | PyTorch | TensorFlow | ROS | Intel OpenVINO | Sci-kit Learn | MLFlow | Optuna | Github | Docker | Jira | CVAT | Matplotlib | Shell Scripting (Linux) | Git

CERTIFICATIONS

Deep Learning Specialisation – 5-course specialisation at deeplearning.ai

AWARDS

Received the **Best Speaker award** on three occasions from ABC Toastmasters club in Muscat, Oman.